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Roll no.

Degree: B.Tech. (ECAM) Semester: 3rd — Course work

MID-SEMESTER EXAMINATION, September, 2024

Course Title: Probability Theory and Random Process

Course Code: **EAEP303 / EAEP306**

Duration: 1:30 Hours

Max. Marks: 20 (EAEP303) / 25(EAEP306)

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer. All questions carry equal marks.

Q. No.	Question	Marks	CO																
1a	Describe random variable, the cumulative distribution function (CDF) and the probability density function (pdf). Discuss the relation between the CDF and the pdf.	2 / 2.5	CO1																
1b	Let Y be a continuous random variable with the probability density function (pdf) given as $f(y) = \text{Exp}(- y)$ Find out (i) $P(-1 \leq Y \leq 1)$ (ii) $\text{Variance}(Y)$	2 / 2.5	CO2																
2a	Consider two discrete random variables X and Y with the following joint probability distribution: <table border="1" style="display: inline-table; vertical-align: middle;"> <tr> <td>P(X, Y)</td><td>Y=0</td><td>Y=1</td><td>Y=2</td></tr> <tr> <td>X=0</td><td>0.1</td><td>0.2</td><td>0.1</td></tr> <tr> <td>X=1</td><td>0.1</td><td>0.1</td><td>0.1</td></tr> <tr> <td>X=2</td><td>0.1</td><td>0.1</td><td>0.1</td></tr> </table> (i) Find marginal distributions of X and Y (ii) Find $E(XY)$	P(X, Y)	Y=0	Y=1	Y=2	X=0	0.1	0.2	0.1	X=1	0.1	0.1	0.1	X=2	0.1	0.1	0.1	2 / 2.5	CO2
P(X, Y)	Y=0	Y=1	Y=2																
X=0	0.1	0.2	0.1																
X=1	0.1	0.1	0.1																
X=2	0.1	0.1	0.1																
2b	The pdf of a random variable X is given by $f_X(x) = 1$ for $0 \leq x \leq 1$, and zero elsewhere. Calculate its variance.	2 / 2.5	CO2																
3a	Find characteristic function of Poisson distribution	2 / 2.5	CO3																
3b	Describe Gaussian distribution. Derive its mean.	2 / 2.5	CO2																
4a	Let $Z = X + Y$, where X and Y are independent random variables, find out the PDF of Z in terms of PDFs of X and Y.	2 / 2.5	CO2																
4b	Explain conditional distribution and conditional expectation.	2 / 2.5	CO2																
5a	Let $Y = aX + b$, X and Y are random variables, a and b are the constants. Derive the PDF of Y in terms of the PDF of X.	2 / 2.5	CO2																
5b	Two random variables- X and Y are independent. Prove that they are uncorrelated.	2 / 2.5	CO3																